

USER MANUAL

NEG POWER MINI

NEG Pump Controller



saes

SAES Getters S.p.A. – Italy
www.saesgroup.com

The **NEG POWER MINI - NEG Pump Controller** power supply serves for operating the **NEG** (Non Evaporable Getters) pumps. It is designed to be used indoor in laboratory conditions.

WARNING

After transportation, the device has to be left idle without mains voltage for at least 3 hours at the ambient temperature.

WARNING

The use of the instrument for any other purpose than the one originally intended may damage the instrument, injure the user and void all warranties.

Safety provisions

ATTENTION

High voltage is present inside the instrument and also in the connectors and cables, which may be harmful to the user.

Manipulation of the cables and of the grounding wire is prohibited while the appliance is in operation.

Likewise, the operation of the instrument without its protective covers is prohibited.

Protect the device against humidity and against penetration of conductive objects and liquids into the ventilation slots.

Symbols on the Manual

These symbols appear on the product:



CAUTION

Pay specific attention
to the section

INDEX

1. APPLICATION.....	4
2. INTRODUCTION	4
2.1. General description	4
2.1. Main Features	5
2.2. Electronic circuit features	5
2.3. Remote and multi Control.....	5
2.4. The front panel of the unit	6
2.5. The rear panel of the unit	6
2.6. NEG POWER MINI Dimensions and Weight.....	7
2.7. Supplied accessories	7
3. INSTALLATION	8
3.1. Rack mounting	8
3.2. Electrical connections	8
3.3. Cleaning	9
4. USAGE.....	10
4.1. Initial setup	10
4.2. Setting Menu	12
4.3. Maximum Allowed Pump Temperature	13
4.4. Logging	13
5. ABNORMAL SITUATION AND ALARM NOTIFICATION	14
5.1. INTERLOCK Alarm	14
5.2. PUMP OPEN Alarm	15
5.3. OVER CURRENT Alarm	15
5.4. PUMP OVER TEMPERATURE Alarm	16
5.5. VMON - UNDER VOLTAGE - OVER VOLTAGE	16
6. RECOMMENDED SETTINGS	17
6.1. Voltage level for the correct heating	17
6.2. Slow Heating	19
6.3. Heating without automatic stop	19
7. THE OUTPUT CONNECTOR FUNCTION.....	20
7.1. Pin layout of DE9 IN/OUT Interface connector.....	20
8. ELECTRICAL SPECIFICATIONS	21
9. MAINTENANCE	21
10. PRODUCT CONFIGURATIONS AND ACCESSORIES.....	22
10.1. NEG POWER MINI configurations and accessories:	22
10.2. Pump cables and accessories:.....	22
11. DECLARATION OF CE CONFORMITY	23
12. INSTRUCTION FOR INSTRUMENT DISPOSAL.....	23
13. WARRANTY CONDITIONS.....	23
14. SERVICE.....	24
14.1. Sales & Service Locations:.....	24
APPENDIX A: Supply cables	25

1. APPLICATION

NEG POWER MINI - NEG Pump Controller power supply is designed to operate the MK4 and MK5 NEG pump series, namely GP SORB AC®, CapaciTorr®, CapaciTorr HV®, NEXTor®, NEXTor HV® and NEXTor® StarCell families.

2. INTRODUCTION

2.1. General description

NEG POWER MINI is an electronic unit specifically designed to control the heating and the activation process of the getter pumps whether they are or aren't equipped with a thermocouple.

NEG POWER MINI can work in the low range of power with the smallest SAES NEG pumps requiring low values of voltage / current / power with a precision around ± 0.1 V.

NEG POWER MINI can control one getter pump at a time, with an adjustable DC voltage output up to 35V, current up to 5A, power up to 175W.

The controller is equipped with Ethernet and RS232 interfaces (to remote control the device) and USB data interface (to record the device parameters on a USB flash memory).

The controller case is a sub-rack 3U 28TE 230mm deep (output connector on the back takes 30 mm of additional space, room for male connector should also be accounted for).

For the local use, it has an alphanumeric display to see the pump and the instrument parameters and four buttons to scroll the menus and modify the parameter values.

It is also possible to control the NEG POWER MINI by the SAES remote control software (this software can control multiple control units, and thus multiple pumps, at the same time).

During the working, the voltage and current output values are shown on the display (together with the pump temperature if available). The voltage value, current value and pump temperature values are provided through the Ethernet and serial communications, and they are also automatically stored in USB flash memory if it is inserted in its slot.

The NEG POWER MINI has a security signal (INTERLOCK) to provide voltage to the pump:

the INTERLOCK contact is necessary to provide voltage to the output and, therefore, it allows to shut down the output voltage to the pump with an external alarm signal.



2.1. Main Features

The main features of the new NEG POWER MINI are the following:

- activation/conditioning/working for NEG pumps;
- adjustable DC voltage output;
- pump temperature monitoring;
- very simple use by local control ;
- remote control via RS232 interface with Modbus protocol (RTU);
- remote control via TCP/IP network interface with Modbus protocol (TCP);
- diagnostic features (open/short connection, broken thermocouple, broken heating filament, overheating control, ...);
- pump temperature limiter: for pumps equipped with a thermocouple, a limit on the maximum temperature can be set by limiting the maximum power.

2.2. Electronic circuit features

NEG POWER MINI is a switching power supply, with an adjustable voltage output from 0 to 35V. The instrument receives mains voltage (100÷240 VAC, 50÷60 Hz), converts it to DC voltage and supplies the getter pump by means of a microprocessor-based electronic control card.

The unit front side has an alphanumeric display and 4 buttons that allow a user to interact.

The unit has, on the back, the instrument input for mains power supply IEC 10A plug with fuse, an ON/OFF button to turn on/off the instrument, one OUTPUT pump connector, a fan, IN/OUT interface connector with INTERLOCK contacts and RS232 serial interface, LAN connector for remote control by standard network interface and the USB data interface (only for USB flash memory logging).

External interlock alarm (alarm condition generated by the user's vacuum system) is present, or the provided bypassing DE9 connector can be used (have to be connect in the IN/OUT interface connector).

If the getter pump is equipped with a thermocouple, **NEG POWER MINI** shows its temperature on the display.

2.3. Remote and multi Control

Dedicated software is provided to remotely control the NEG POWER MINI through the Ethernet connection. The software allows controlling several NEG POWER MINI units at the same time.

All functions and parameters can be easily controlled through the software.

Software allows controlling a single unit or more units at the same time.

Local and remote control are possible contemporarily.

This software is available for Windows and for Linux PCs.

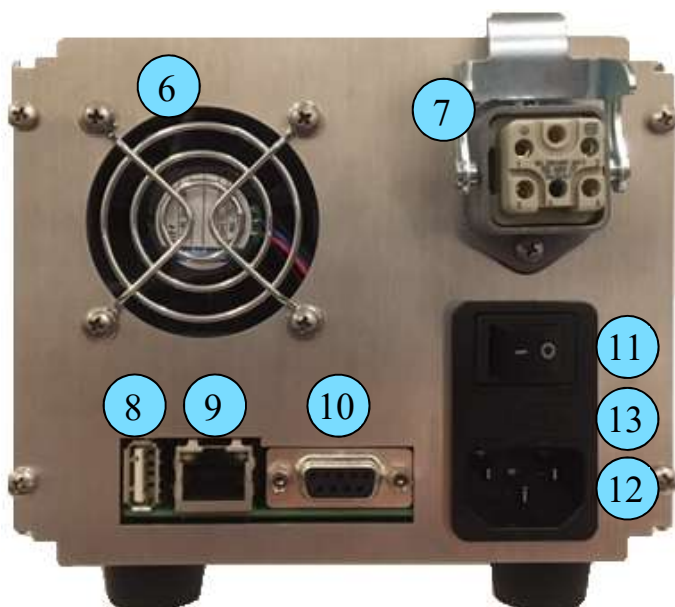
The communication protocol via Ethernet and RS232 is MODBUS (TCP and RTU) (for more details see the dedicated paragraph).

2.4. The front panel of the unit



1. Alphanumeric display;
2. MENU/ENTER button to edit the output parameters and to enter the setup menu;
3. ▽ to reduce the selected value / scroll down in the menu;
4. ▲ to increment the selected value / scroll up in the menu;
5. START/STOP button to switch on/off the output.

2.5. The rear panel of the unit



6. Fan for instrument cooling;
7. OUTPUT pump connector;
8. USB connector (to record the device parameters on a USB flash memory);
9. Ethernet connector (to remote control the device);
10. IN/OUT interface connector with INTERLOCK and RS232 connector;
11. ON/OFF instrument power switch;
12. instrument mains plug;
13. fuse (T3.15AL250V)

2.6. NEG POWER MINI Dimensions and Weight

- NEG POWER MINI dimensions 28TE 19" sub- rack, 3U, depth 230 mm
- NEG POWER MINI weight 1.5 kg

2.7. Supplied accessories

The following accessories are provided with NEG POWER MINI:

- Mains power cable with C16 plug and Schuko plug;
- DE9 plug with PIN to bypass Interlock connections;
- Saes Group USB drive with:
 - o User manual;
 - o NEG POWER MINI manager: software for remote control via Ethernet.

3. INSTALLATION

The unit may be installed on a desk or mounted in a rack. Place the instrument away from heat sources. A gap of at least 5 cm must be left between the sides of the unit and adjacent walls to ensure sufficient air flow to the cooling fan. If the NEG POWER MINI is installed inside a cabinet, the latter must be provided with vents capable of ensuring adequate air exchange between the inside and the outside.

 **ATTENTION:** The device must be positioned / installed in such a way that it can be easily disconnected from the mains if needed.

3.1. Rack mounting

Insert the instrument in the appropriate space (19" rack 3U according to DIN 41494 standard).
Fix the NEG POWER MINI front panel on the rack with four screws.

 **ATTENTION:** the instrument cannot be used in potentially flammable or explosive areas.

 **ATTENTION:** do not use accessories which are not supplied or recommended by the manufacturer.

 **ATTENTION:** to avoid damages of the electronic circuit use the instrument within the environment specified conditions.

3.2. Electrical connections

The connections for the pump, the mains line, the interlock and the external interface are on the rear panel of the instrument.

Before connecting the unit to the mains supply, make sure that supply voltage is within the range specified in the section Electrical Specifications.

 **ATTENTION:** protective earthing is required!

Plug the equipment into a supply outlet which has an earth connection.

 **ATTENTION:** use only adequate rated cords to connect instrument to the mains supply.

 **ATTENTION:** do not use cables which are not supplied or recommended by the manufacturer.

 **ATTENTION:** in order to disconnect the NEG POWER MINI from the mains, turn off the power switch first and then unplug the Schuko plug from the mains.

3.3. Cleaning



ATTENTION: before starting any cleaning turn the unit off and remove all mains.

The units should be cleaned with a damp cloth when necessary.

The use of solvents or of an aggressive or scouring cleaning agent is not permitted.

Avoid spraying or spilling any type of liquid directly onto the unit.

Before putting the unit into operation mode, check that it is thoroughly dry.

4. USAGE

NEG POWER MINI has an alphanumeric display and four buttons for the local control; it is also possible to control the NEG POWER MINI by the SAES remote control software or by a customer remote command sent in Modbus communication protocol



ATTENTION: Before starting to heat the getter pump, check it is under high vacuum level.

If heated in air, the getter pump could get damaged.

4.1. Initial setup

Before switching on the instrument, connect the power supply cable, the output cable towards the pump and the interlock enable signal (or pre-cabled DE9 connector provided with the unit).

The unit is switched on by setting the **I-O** switch into position **I**. The display turns ON.

Note: The **NEG pump** is not electrically powered when the supply is switched on.

The first screen shows the name of the instrument and the software version.



After few seconds, the display changes to the **Standby** screen identified by the word "Standby" on the first line.



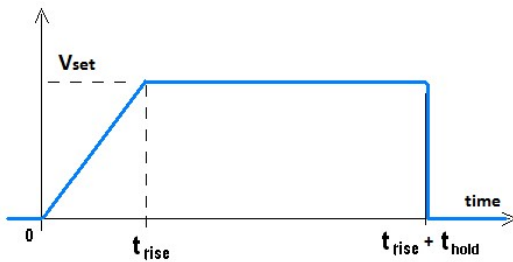
The second line reports the current temperature read by the pump thermocouple (if equipped) or three dashes that indicate the missing reading usually due to the fact the thermocouple is not available in the pump.

Third line shows the 3 parameters that the user can set and, in the fourth line, the currently set values are visible.

Vset is the voltage that will be applied to the pump.

Trise is the time in minutes that the unit will take to ramp the voltage up from 0 to Vset

Thold is the time until self switching off, that is the time interval during which the output shall be at **Vset** before switching off the power to the pump. It is possible not to give a value to **Thold** (three dashes shown) which means that no self switching off will take place (the user will have to stop the powering by acting on START/STOP button).



Factory default setting is **Vset** = 0.0V **Trise** =15' **Thold**= --- (not set)

In order to modify setup values, press shortly MENU/ENTER.

A blinking cursor will indicate which parameter is currently being edited.

By shortly pressing arrow keys you will increment or decrement by 1 the value. A long pressure on an arrow key will change the edited value fastly.

Pressing MENU/ENTER is the action to move to the next parameter. After the last value pressing MENU/ENTER saves the values.

START/STOP activates and stops the power supply: pressing the button once starts supply and pressing it again stops the supply at any time.

START/STOP is not enabled while editing parameters.

When you turn off and turn on NEG POWER MINI, the last settings and used parameters are presented and you can use them again pressing start button without any other change.

So it is easy to choose the settings one time and use repeatedly on the same system; if you change system or usage conditions you can choose the new appropriate settings.

During power supply to the pump, the **Run** screen is shown.



The second line reports the **temperature** of the pump (if available).

The third line shows the value of **Vset** as chosen in the Setup menu and alongside **Ton** indicating the number of minutes from the beginning of the power supply to the pump.

Fourth line identifies the current output supply condition "**Out**" and we can read both voltage and current.

After the end of power supply, whether obtained by pressing STOP or by self switching off, the **Standby** screen is shown again.

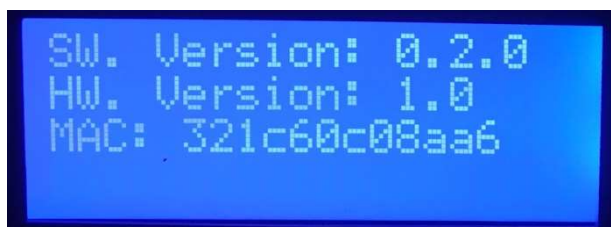
In the Run screen it is possible to change **Vset** value without interrupting output supply: this result can be achieved by keeping pressed MENU/ENTER and acting at the same time on the arrow keys. The new value will be active as soon as MENU/ENTER is released.

Please note that this value replaces the setting and after the end of current supply the **Vset** in the Standby page will be also updated with the new value.

4.2. Setting Menu

With a long press on MENU/ENTER it is possible to enter in the pages to configure the instrument parameters.

The pictures show the default parameters of NEG POWER MINI.



The first (read only) page shows the software version, the hardware version and the MAC address. With the down arrow you can scroll to the second page.



Here MODBUS TCP can be enabled and IP parameters can be changed.

A short pressing on MENU/ENTER activates editing of the first field and moves to the next field, arrow keys change parameter values. After editing the last parameter (Port) the next pressing on MENU/ENTER ends editing.

The following parameters are available:

Enable/Disable: Enable/Disable the Modbus over TCP/IP on the Ethernet interface

IP: IP address, the four bytes can be edited separately.

GW: the IP address of the gateway, the four bytes can be edited separately.

Mask: is the IP address net mask encoded as the number of bits corresponding to the network portion of the IP address (for example 24 corresponds to 255.255.255.0)

Port is the TCP port where communication takes place.



Third page shows Modbus RTU parameters for Modbus over RS232 (DE9 plug).

A short pressing on MENU/ENTER activates editing of the first field and moves to the next field, arrow keys change parameter values. After editing the last parameter (Slave) the next pressing on MENU/ENTER ends editing.

The following parameters are available:

- Enable/Disable: Enable/Disable the Modbus over RS232;
- Baudrate is the serial port speed in bit/s;
- Parity is the parity bit configuration of RS232;
- Stop is the stop bit configuration of RS232;
- Slave is the slave ID of the instrument.

4.3. Maximum Allowed Pump Temperature



Fourth page is the additional protection parameter for the customer who wishes to use it.

By setting Tpump max, the maximum pump temperature is defined above which the instrument stops supplying power to the pump.

By default it is not set (OFF).

4.4. Logging



If a pen drive with FAT32 formatting is inserted in the USB port, the instrument automatically logs

the parameters during the run phase.

An "L" is shown to indicate that log is being recorded.

The log file is a text file organized in four columns: as described in the first line, the columns are: time from the start, output voltage, output current, pump temperature.

In the second line we find the setting parameter values. The setting values are also reported in the file every time they are changed.

See the following example:

```
# t[s]; Vout[V]; Iout[A]; Tpump[°C];  
# Vset=24.0 Trise=000' Thold=000'  
0;23.9;5.2;26;  
10;24.0;5.2;26;  
20;24.0;5.1;28;  
30;24.0;5.1;31;  
40;24.0;5.1;35;  
50;24.0;5.1;39;  
60;24.0;5.0;45;  
# Vset=26 Trise=000' Thold=000'  
70;24.5;5.0;52;  
80;26.0;5.1;60;  
90;26.0;5.3;69;  
100;26.0;5.3;79;
```

5. ABNORMAL SITUATION AND ALARM NOTIFICATION

If there are problems during the control of NEG pumps, the **Alarm** screen is shown.

Press the ENTER button to acknowledge the alarm and the **Standby** screen is shown again.

The most common situations are the following:

5.1. INTERLOCK Alarm



This notification indicates that the interlock signal is open.

As we said before INTERLOCK (close contact) is an enable signal, so it is necessary to close short interlock pins to provide power to the pump. You can provide the close contact with an external circuit or with the pre-cabled DE9 connector provided with the unit.

5.2. PUMP OPEN Alarm



This notification indicates that the pump is not connected.

Please check that the cable is well connected to both the pump and the instrument.

If the alarm persists, please contact SAES GETTERS S.p.A. for assistance.

5.3. OVER CURRENT Alarm



This notification happens when the output current is permanently over 5A.

It indicates that the electrical setting is too high for that pump model.

Please, check on the pump manual the recommended voltage setting for your pump model and your pump installation configuration, reduce the voltage setting in order to try again and contact SAES GETTERS S.p.A. for any assistance you need.

5.4. PUMP OVER TEMPERATURE Alarm



If you are using the setting T_{pump max} (the maximum pump temperature allowed) this notification indicates that the pump temperature exceeded the setting value and the instrument stops supplying power to the pump (here for example 260°C)

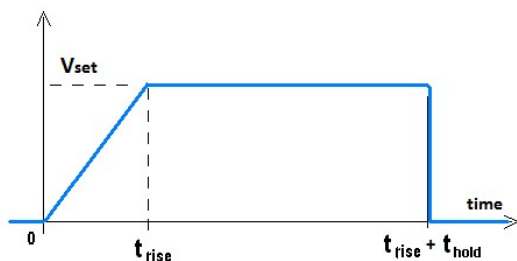
5.5. VMON - UNDER VOLTAGE - OVER VOLTAGE

Other possible alarm situations are VMON or UNDER VOLTAGE or OVER VOLTAGE.

These notifications indicate that there are some problems with the internal circuit and/or with the internal reference voltage levels. Please, contact SAES GETTERS S.p.A. for assistance.

6. RECOMMENDED SETTINGS

As explained in chapter 4 “USAGE” voltage to the pump is supplied according to the following time graph.



where **Vset**, **t_rise** and **t_hold** are parameters that can be chosen by the user.

When the instrument is switch on the presented parameters are the last set during previous session,

Before switching the instrument on, you should make sure that the parameters are suitable for the pump you intend to control in this specific session.

6.1. Voltage level for the correct heating

In the manual of each SAES pump you can find detailed instruction on how to operate your NEG pump in the best way, the optimal target temperature and electrical conditions to reach it in all different configurations where the pump can be installed; moreover you will find information about installation and the impact on heating and required powering.

A general indicative rule is to apply the maximum voltage during activation of the pump in the nude configuration (without jacket).

If the pump is with jacket, voltage to be applied should be reduced by 10-20% (this changes according to the model and nipple size).

During conditioning a voltage of about 50% with respect to activation should be used.

If the pump is equipped with a thermocouple, you should always rely on the measured temperature.

If the target temperature cannot be achieved due to the system limitations, an increase of the activation time should be foreseen.

The default values of Voltage, Current, Power and Temperature are the maximum recommended settings for the selected pump.

They are recommended / necessary for pumps without jackets and without heat source nearby. If the system includes a partial or a full jacket around the pump, a lower power is required to reach the correct Temperature. For more details refer to the pump manual.

It is always possible to regulate Voltage value in the setup menu.



ATTENTION: SAES Getters is not responsible for damages due to improper use of the power supply

In the following table, for the example, the temperature and the corresponding voltage current and power parameter settings are reported for activation and conditioning of the most popular SAES getter pumps, like the CapaciTorr and the NEXTorr families, without jacket.

NAME		Activation Mode	Conditioning Mode
		Vmax	Vmax
CT D100		9 V	3.5 V
CT D200		12 V	4 V
CT D400-2		16 V	7 V
CT D50		8 V	3 V
CT HV 200 35		11.3 V	3.3 V
CT HV 200 63		12.6 V	3.7 V
CT Z100		8 V	4 V
CT Z200		11 V	5 V
CT Z400 body		15 V	5 V
CT Z400 nude		18 V	7 V
GP50-2F		25 V	11 V
NXT D100-5		9 V	4.5 V
NXT D200-5		12.5 V	6 V
NXT D300-5		20 V	8 V
NXT D500-5		28 V	10 V
NXT HV 100		9 V	2.8 V
NXT HV 200		12 V	3.5 V
NXT HV 300		30 V	8 V
NXT Z100		9 V	4 V
NXT Z200		12 V	5 V
NXT Z300		24 V	9 V
NXT D500-StarCell		28 V	10 V



ATTENTION: SAES Getters is not responsible for damages due to improper use of the power supply

6.2. Slow Heating

Trise is the time that the unit will take to ramp the voltage up from 0 to Vset. Factory default is 15'.

If activation starts at room temperature and in general each time when possible, it is recommended to use always a long Trise to heat the pump gradually. When supply start at an already high temperature or in case fast heating is needed it is possible to reduce Trise down to 0 (in this case Vset will be reached in a few seconds).

Once Vset is reached, Vset voltage is kept for Thold.

6.3. Heating without automatic stop

Some applications require special getter pumps (HV) that work when the getter material is hot. NEG POWER MINI can be used to keep these pumps hot giving them the necessary power continuously. To do this, t_hold = --- (not set) is used.

(to set t_hold = --- is the same as to put it = 0).

When t_hold = --- (not set) the power supply is only stopped by pressing the stop button.

t_hold = --- (not set) can also be used with standard NEG pumps when the pump need heating for a very long time.

7. THE OUTPUT CONNECTOR FUNCTION

The output connectors can be used for a possible connection of a device designed for monitoring or controlling the supply activity.

IN/OUT INTERFACE is intended for connection to protective hardware systems;

RS232 connector can be used for communication purposes via RS232 interface in MODBUS RTU protocol.

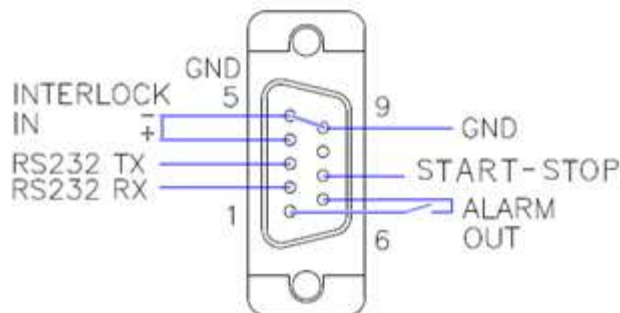
LAN connector can be used for communication purposes via ETHERNET interface in MODBUS TCP protocol.

(Modbus Application Protocol V1.1b3)

7.1. Pin layout of DE9 IN/OUT Interface connector

Interlock couple wires are open / short inputs. Short shall sustain at least 5 mA, open will present a 12V voltage difference.

Alarm couple wires are open / short outputs. They can be read with a maximum voltage of 24 Vdc and a maximum current of 500 mA.



8. ELECTRICAL SPECIFICATIONS

Input:

Maximum Power	250 W
Supply voltage	100-240 V~, +/-10%
Frequency	50/60 Hz +/- 10%
Input Current	2.5 A max
Mains cord connector	IEC Type 10A 250V
Over-Voltage category	II
Pollution degree	2
Isolation type	class I
Altitude max	up to 2000 m
Operating Temperature	0°C to 40°C
Relative Humidity	80%
Noise at 1 mt	< 40 dBA

Output:

Output power	175 W + overload
Output Voltage	0-35 V $\Rightarrow$
Supply Current	5 A $\Rightarrow$
Overload	110%

9. MAINTENANCE

The supply needs minimum maintenance. Depending on the dust levels in the environment where the device is being operated, a periodic check of the device is desirable. In case of dust contamination it can be cleaned by means of a vacuum cleaner.

Should the NEG Power Mini be repaired, it must be sent back to SAES Getters S.p.A..

10. PRODUCT CONFIGURATIONS AND ACCESSORIES

Configurations and accessories are given in the following section:

10.1. NEG POWER MINI configurations and accessories:

<u>Product description</u>	<u>Code</u>	<u>Note</u>
NEG POWER MINI	3B0110	Controller for NEG Pumps
Main Input Cable	3B0338	Main cable with IEC 10 A plug and Schuko
Interlock connector		DE9 connector with interlock connections

10.2. Pump cables and accessories:

Product description	Code	Pump Models
NEG cable 3P5A 3MT ^{*§}	3B0598	Capacitorr D 50, D 100, D 200, Z 100, Z 200; NEXTorr D 100, D 200, Z 100, Z 200, HV 100, HV 200.
NEG cable 4P10A 3MT ^{*§}	3B0601	Capacitorr D 400, Z 400, HV 200.
NEG cable 6P5A 3MT ^{*§}	3B0854	NEXTorr D 300, D 500, Z 300, Z 500, HV 300.
NEG cable for GP50 3MT [*]	3B0370	GP 50.

(*) Other length cables are available on request

(§) Bakeable cables up to 250 °C, and radiation resistant (1000 Mrad)



ATTENTION: do not use accessories which are not supplied or recommended by the manufacturer.

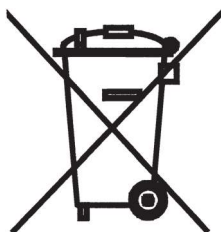
11. DECLARATION OF CE CONFORMITY

(see document attached)

12. INSTRUCTION FOR INSTRUMENT DISPOSAL

The instrument disposal has to be carried out in compliance with the user's country applicable regulations.

The information below is issued in compliance with the regulations as set out by the 2012/19/EU directive (Waste Electrical and Electronic Equipment).



The instrument contains materials which may endanger the environment and it's not allowed to dispose it with unsorted urban waste.

The equipment shall also be disassembled by material for disposal.

The different materials shall be collected separately in accordance with local waste disposal legislations.

Neither the collection nor the transport of thus collected and separated materials is subject to any special requirements.

13. WARRANTY CONDITIONS

SAES guarantees that the Products delivered shall be free from operational and material defects and shall comply with the construction and functional data and specifications indicated in the Contractual Documents.

This warranty shall have a term of TWELVE (12) MONTHS. For Products which require installation at BUYER's facility by SAES personnel, the warranty shall have a term of TWELVE (12) MONTHS from the date of installation or FOURTEEN (14) MONTHS from the date of delivery, whichever term is shorter. Subject to the remainder of this Article 14, any action by BUYER for any alleged breach of this warranty shall be brought in writing by BUYER within thirty (30) days of BUYER's discovery of the breach. This warranty shall only apply to the BUYER and may not be assigned.

During the term of the warranty set forth above, SAES will promptly repair the Products which for their features can be repaired and which do not conform to the specifications and which BUYER returns to SAES at the address provided. Unless otherwise agreed and specified, BUYER shall be responsible for all transportation charges incurred in returning Products to SAES for repair; BUYER shall have obtained a Returned Material Authorization ("RMA") number and specific shipping instructions from SAES prior to its shipping of the Products to SAES. SAES shall not unreasonably deny BUYER authorization to ship Products to SAES. SAES shall return repaired Products to BUYER, with transportation charges prepaid by SAES, unless otherwise agreed. Additional information is available on the General conditions of sales.

14. SERVICE

For a request of return of the component contact a SAES Customer Service and will receive a **Return Merchandise Authorization (RMA)** number.

Please note that SAES cannot accept any instrument which contains biological or chemical hazards or radioactive substances. Please clearly inform SAES Customer Service should this have happened during instrument use, in order to discuss adequate solutions.

14.1. Sales & Service Locations:

Europe, Middle East and Africa:

SAES Getters S.p.A.

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20045 Lainate (Milan) - Italy
Ph. +39 02 93178 1 - Fax +39 02 93178 320

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E-mail: CRM_SALES@saes-group.com

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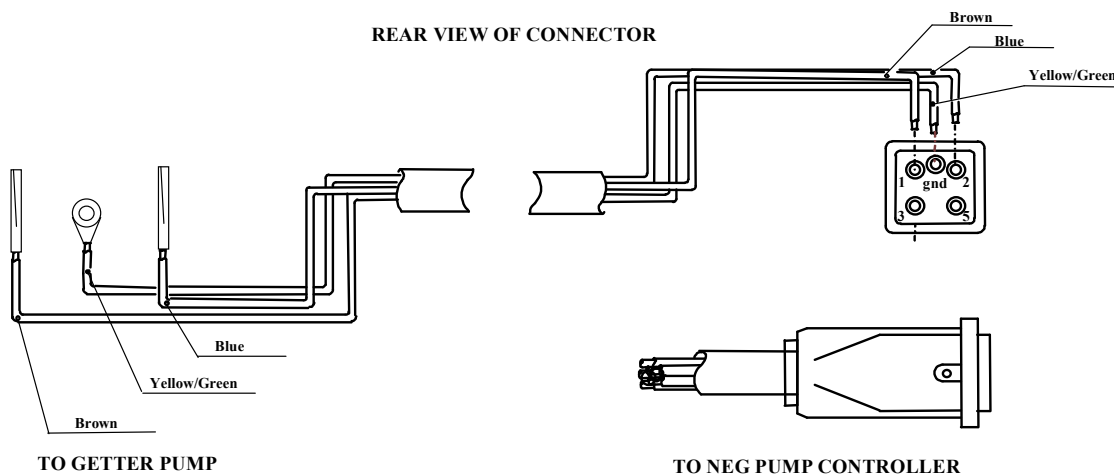
SAES Getters USA, Inc.

1122 East Cheyenne Mountain Blvd.
Colorado Springs, CO 80906 - USA
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APPENDIX A: Supply cables

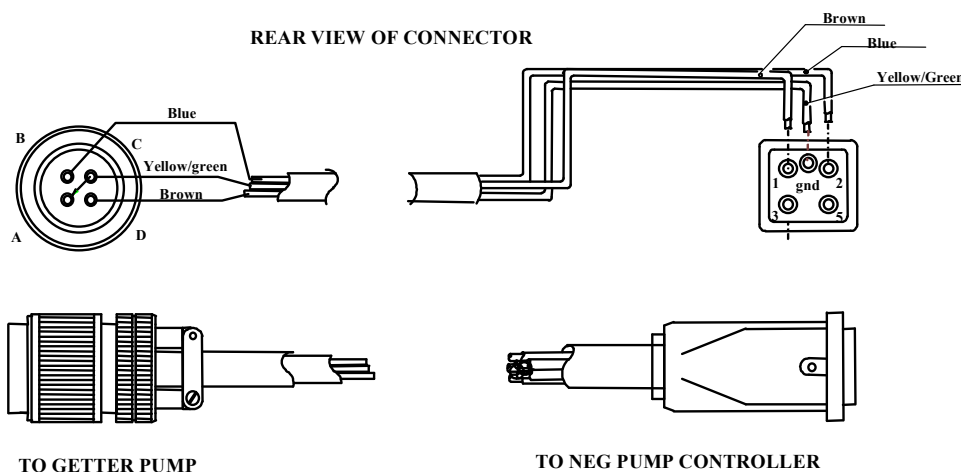
GP 50-2F

For the GP 50 pump and pumps not equipped with an electrical connector use the output supply cable supply shown in figure:



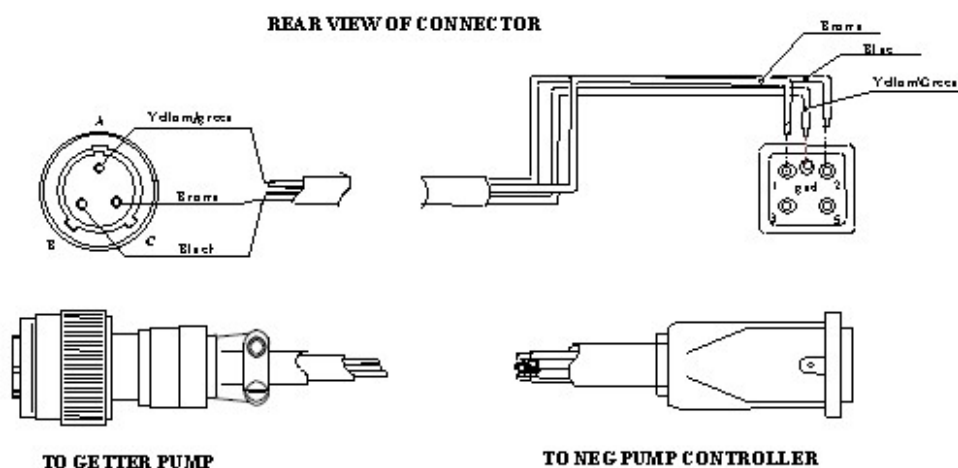
CapaciTorr D400-2, CapaciTorr Z400 CapaciTorr HV200 35, CapaciTorr HV200 63

For these pumps which are equipped with a dedicated electrical connector, use the output supply cable shown in figure:



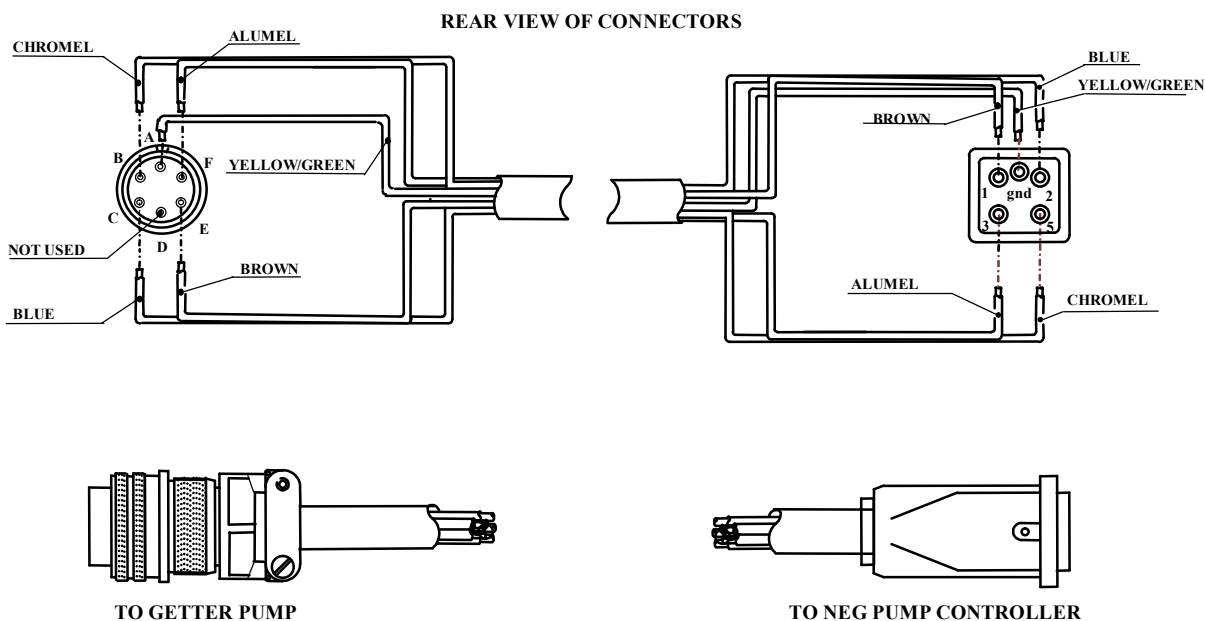
**CapaciTorr D50, CapaciTorr D100, CapaciTorr D200,
NexTorr D100-5, NexTorr D200-5, NexTorr HV 100, NexTorr HV 200,
NexTorr Z100, NexTorr Z200, CapaciTorr Z100, CapaciTorr Z200,**

For these pumps which are equipped with a dedicated electrical connector, use the output supply cable shown in figure:



**NexTorr D300-5, NexTorr D500-5, NexTorr HV 300, NexTorr Z300
NexTorr D500-StarCell**

For these pumps which are equipped with a dedicated electrical connector, use the output supply cable shown in figure:





SAES Getters S.p.A. – Italy
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